

Integrate HBase and Hive Tables and show how both are sync with each other.

Integrating **HBase and Hive** allows you to **query HBase tables using HiveQL**, giving the flexibility of Hive's SQL-like interface with the NoSQL scalability of HBase.

Below is a **step-by-step guide** to integrate **HBase and Hive**, and demonstrate how they can stay **in sync** with each other.

Step-by-Step Integration: Hive + HBase

Step 1: Start All Required Services

start-dfs.sh

start-yarn.sh

start-hbase.sh

hive

Step 2: Create an HBase Table

In the HBase shell:

hbase shell

create 'student', 'info'

put 'student', '1', 'info:name', 'Amit'

put 'student', '1', 'info:dept', 'IT'

put 'student', '2', 'info:name', 'Neha'

put 'student', '2', 'info:dept', 'CS'

scan 'student'

Step 3: Create a Hive Table Mapped to HBase Table

Now switch to Hive:

CREATE TABLE hbase_student (

```

    key STRING,
    name STRING,
    dept STRING
)
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
WITH SERDEPROPERTIES (
    "hbase.columns.mapping" = ":key,info:name,info:dept"
)
TBLPROPERTIES (
    "hbase.table.name" = "student"
);

```

Step 4: Query the Hive Table (That Pulls Data from HBase)

```
SELECT * FROM hbase_student;
```

Output:

```

1  Amit  IT
2  Neha  CS

```

Step 5: Insert Data into Hive Table

Let's insert a new row using Hive:

```
INSERT INTO TABLE hbase_student VALUES ('3', 'Ravi', 'ECE');
```

Now check from **HBase shell**:

```

hbase shell
scan 'student'

```

Output in HBase:

```

ROW  COLUMN+CELL
3    column=info:dept, value=ECE

```

3 column=info:name, value=Ravi

Sync Between HBase and Hive

Action	Performed In	Reflected In
Insert/Update	HBase	Queryable in Hive
Insert (Hive)	Shows in HBase	Automatically
Scan (HBase)	Shows Hive Insert	Yes

Conclusion:

The **Hive table acts as a bridge** to the HBase table, allowing bidirectional updates and real-time sync, as long as the schema and column mappings are correctly defined.